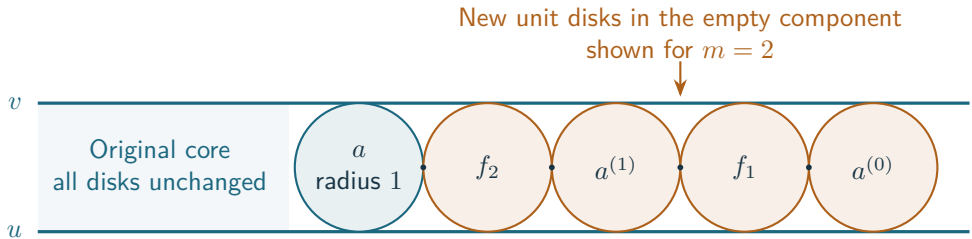
Add twist regions inside the empty marked gap



$D^{(m)} = (1, 2)^m D$ in zero-based slots

$2m$ new boxes, each with an independent even signed exponent

Every old curvature stays fixed; each new curvature is 1.

The first old input middle changes $0 \to 1$, increasing its balance margin.

Any balanced certified core starting in slot 1, using no slot 0,
with its first cap marked. The twist threshold may depend on m.